Yutong Wang

CONTACT INFORMATION

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RESEARCH INTEREST

I am a second-year PhD student from the University of Sydney. My research interests mainly focus on machine learning, especially **multi-modal learning, optimal transport, video understanding and generation**.

EDUCATION

PhD, Computer Science
The University of Sydney, NSW, Australia — Sep. 2025–Present
Advisor: *Prof.* Chang Xu

MPhil, Computer Science
Beijing Institute of Technology, Beijing, China — Sep. 2022–Jun. 2025
Advisors: *Prof.* Dixin Luo, *Prof.* Hongteng Xu

Bachelor, Computer Science
Beijing Institute of Technology, Beijing, China — Sep. 2018–Jun. 2022

INTERNSHIP EXPERIENCE

1. **Shanghai AI Laboratory**, *Research Intern* — Apr. 2025–Present
   - Explore the application of VLM and reinforcement learning to achieve **graphic design tasks**, such as poster design, cover design, and web design.
   - Explore optimal transport distillation techniques to **accelerate video generation**, have achieved acceleration for the VACE model, enabling efficient video generation in just 4 steps. Also, explore content-adaptive block routing techniques for DiT depth pruning and spatial-temporal aware width pruning, which further nearly halves per-step computation for both I2V and T2V generation tasks.

   Advisors: Xinyuan Chen, Yaohui Wang

2. **Ant Group**, AliPay, *Research Intern* — Apr. 2024–Oct. 2024
   - Participate in the audio-driven portrait animation project EchoMimic. My research topic is **video face enhancement**, including blind face video restoration and de-flickering in real-world and AI-generated videos.
   - Proposed a novel **3DVQGAN-based video enhancement framework with a powerful discriminator** that can support effective compression.
   - Proposed **spatial-temporal codebooks** for video quantization and **marginal prior regularization** to mitigate the codebook collapse problem.

   Advisors: Jiajiong Cao, Chenguang Ma

3. **VRC Inc.**, *Research Intern* — Jul. 2023–Sep. 2023
   - The research topic is **stylized video generation** based on text descriptions or referenced images, focusing on VLOG videos.
   - Proposed a stylized video generation pipeline, that can achieve DreamBooth-based style transfer, SD-based quality enhancement, and video stability enhancement (de-flickering).

Advisors: Hongteng Xu, Yingdi Xie

PUBLICATION CONFERENCE

1. **Yutong Wang**, Yunke Wang, Tianfan Xue, Yu Qiao, Yaohui Wang, Xinyuan Chen, Chang Xu - PARE: Pruning and Adaptive Routing for Efficient Video Generation, International Conference on Neural Information and Processing Systems (**NeurIPS**), 2026.
2. Jiaxin Zhang, Dixin Luo, Jiayi Liang, **Yutong Wang**, Bohan Zhang, Yingdi Xie, Hongteng Xu - RobustVTON: Robust Virtual Try-on Benefited from Debiased Model Adaptation, The 9th Chinese Conference on Pattern Recognition and Computer Vision (**PRCV**), 2026.
3. **Yutong Wang**, Haiyu Zhang, Tianfan Xue, Yu Qiao, Yaohui Wang, Chang Xu, Xinyuan Chen - VDOT: Efficient Unified Video Creation via Optimal Transport Distillation, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
4. Sidan Zhu, **Yutong Wang**, Hongteng Xu, Dixin Luo - Weakly-Supervised Movie Trailer Generation Driven by Multi-Modal Semantic Consistency, The International Joint Conference on Artificial Intelligence (**IJCAI**), 2025.
5. **Yutong Wang**, Jiajie Teng, Jiajiong Cao, Yuming Li, Chenguang Ma, Hongteng Xu, Dixin Luo - Efficient Video Face Enhancement with Enhanced Spatial-Temporal Consistency, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
6. **Yutong Wang***, Sidan Zhu*, Hongteng Xu, Dixin Luo - An Inverse Partial Optimal Transport Framework for Music-guided Movie Trailer Generation, The 32nd ACM International Conference on Multimedia (**ACM MM**), 2024.
7. **Yutong Wang**, Hongteng Xu, Dixin Luo - Self-supervised Video Summarization Guided by Semantic Inverse Optimal Transport, The 31st ACM International Conference on Multimedia (**ACM MM**), 2023.
8. Dixin Luo, **Yutong Wang**, Angxiao Yue, Hongteng Xu - Weakly-Supervised Temporal Action Alignment Driven by Unbalanced Spectral Fused Gromov-Wasserstein Distance, The 30th ACM International Conference on Multimedia (**ACM MM**), 2022.

TECHNICAL REPORT

1. Chengcheng Wang, Yuheng Shi, Siyu Xu, **Yutong Wang**, Jinjing Zhao, and others - Auriga Technical Report: A Task-Agnostic Agent Harness for Zero-Shot Embodied Manipulation, arXiv.

PREPRINT

1. **Yutong Wang**, Yunke Wang, Xinyuan Chen, Chang Xu - BEAT: Rhythm-Elastic Alignment for Agentic Music-guided Movie Trailer Generation, arXiv 2605.27067.
2. Enhuai Liu, Yunke Wang, **Yutong Wang**, Changming Sun, Chang Xu - LoSA: Near-Lossless Sparse Attention for Training-Free Video Diffusion Acceleration, arXiv 2608.12032.
3. **Yutong Wang**, Xingtong Ge, Enhuai Liu, Yunke Wang, Tianfan Xue, Yu Qiao, Yaohui Wang, Xinyuan Chen, Chang Xu - VDOT++: Unified Few-Step Video Generation via Unbalanced Optimal Transport Distillation, arXiv 2610.03221.

4. **Yutong Wang**, Xingtong Ge, Enhuai Liu, Yunke Wang, Tianfan Xue, Xinyuan Chen, Chang Xu - PulseQuant: Propagation-Guided Subspace Correction for 4-Bit Video Diffusion Transformers, arXiv 2609.33384.
5. Xingtong Ge, **Yutong Wang**, Lunjie Zhu, Fangyu Lin, Haitao Lin, Yushi Huang, Xin Zhang, Yi Zhang, Yu Liu, Jun Zhang - Salt++: Context-Aligned Post-Training for Few-Step Streaming Multimodal Generation, arXiv 2609.36995.

* denotes Equal contribution.

PATENT

1. An audio-guided video generation method. CN Patent ZL 202411174397.6
2. A self-supervised method for video summarization. CN Patent ZL 202311104554.1.
3. A Retrieval Method. CN Patent ZL 202211404021.0.
4. A Video Understanding Method. CN Patent ZL 202211405957.5.

TEACHING EXPERIENCE

Tutor, The University of Sydney. Feb. 2026–May. 2026
Course Name: COMP5329 Deep Learning
Instructor: *Prof.* Chang Xu

Teaching Assistant, Beijing Institute of Technology. Sep. 2022–Jun. 2024
Course Name: Programming with Python
Instructor: *Prof.* Dixin Luo

HONOR

USydIS Scholarship, The University of Sydney	2025
Outstanding Graduate, Beijing Institute of Technology	2025
National Scholarship, Ministry of Education of China	2024
Travel Grant Award of ACM MM, MM	2024
Travel Grant Award of ACM MM, SIGMM	2023
Merit Student, Beijing Institute of Technology	2023
Outstanding Graduate, Beijing Institute of Technology	2022

SELECTED TALKS

Efficient Unified Video Creation via Optimal Transport Distillation
Invited talk on Sydney AI Meetup, May 2026.

Generative AI with Diffusion Models
Invited lecturer at USYD GenAI Summer School, Dec 2025.

Efficient Video Face Enhancement with Enhanced Spatial-Temporal Consistency
Invited talk on AI TIME, May 2025.

SERVICE

Reviewer (Conference): ACM MM, ICCV, WACV, CVPR, BMVC, ECCV, NeurIPS, ICLR
Reviewer (Journal): T-MLR, T-IP, T-PAMI